

## 40. Noncommutative Algebras

*Math. Zs. 37 (1933), 514–541*

The main theorems of commutative algebra are, as is known, contained in Galois theory, preceded by the theory of adjunction fields and splitting fields: fields sufficient for a given polynomial to split off one linear factor, or to split completely into linear factors.

Here I develop the corresponding parts of algebra in the noncommutative case, especially in the hypercomplex case. In principle I work with noncommutative methods, namely representation in noncommutative fields. At the end I show how the above-mentioned theorems of commutative algebra can be proved in a completely parallel manner by means of representation in commutative fields.

The underlying representation theory, preceded by a short automorphism theory (§1), is a further development of the theory based on representation modules. In particular, alongside direct representation I consider reciprocal representation. Each can be reduced to the other, but the reciprocal representation is now generated by the reciprocal representation module. The advantage is that the reciprocal representation module, hence a bimodule, can also be regarded as a one-sided module over an extension ring (§2). Thus representation in noncommutative fields is reduced to ideal theory in the extension ring; the irreducible representation classes correspond to the irreducible ideal classes of the extension ring, in exact analogy with the facts which hold for a hypercomplex system itself when represented over its commutative coefficient field (§§3, 4).

From this point one obtains the structural theorems for matrix rings over noncommutative fields by observing (§5) that every subring gives a representation by this field, or equivalently a reciprocal representation by the reciprocally isomorphic field. This reciprocally isomorphic field is a first analogue of a minimal adjunction field and splitting field in the commutative case: it mediates all reciprocal representations of first degree, and, when the rank over the center is finite, also a full decomposition into direct summands of rank 1. This gives the Galois theory for fields (§6) and also expresses the fact (§7) that reciprocally isomorphic division algebras, i.e. fields of finite rank over their centers, generate inverse classes in R. Brauer's group of algebra classes.

A second proof of the Galois theory, valid more generally for simple systems, is almost an immediate consequence of a theorem on commuting subrings, which in turn is connected almost directly with the preceding automorphism considerations. Up to this point the arguments are purely noncommutative. But the question of commutative splitting fields is also treated noncommutatively by representing the splitting field through the division algebra, using the observation prepared in §5 (§7). The final part gives the transfer to the commutative case (§8) and the theory of splitting fields for arbitrary systems (§9).

R. Brauer based the theory of splitting fields on this representation theory in the commutative case, and on irrational factor systems. Because of this commutative foundation he, and later Albert, had to assume the center to be a perfect field, a restriction which is unnecessary in the noncommutative foundation. With the same restriction, again using representation theory in the commutative case, R. Brauer and K. Shoda further developed the theory.

Finally I mention that, for fields, a short account is found in van der Waerden, following my lecture course of summer 1928. Van der Waerden's account introduced several simplifications relative to that course; some of these I adopted in a second course, and some only here. The theorem on commuting subrings (§5) and the second proof of the noncommutative Galois theory based on it (§6, 1 and 2) were added only later. Certain arguments of the first course, the method of forming intersections of difference ideals, have the advantage, though they are more complicated, of transferring to systems of infinite rank and to integral systems. For commutative integral systems this leads to the connection between ideal differentiation and the different.

The principal application of the theory of splitting fields lies in the theory of crossed products and their factor systems, which in turn gives the basis for applications in number theory; this will not be pursued here.

### §1. On Automorphisms, Modules, and Bimodules

Representation theory based on representation modules rests on the theory of the automorphism ring of abelian groups, with or without operators. The implicit arguments involved, namely the relations between mappings and calculation laws, are formulated here as a few simple propositions in order to avoid repetitions.

**1. Multiplicative mapping. Associative law.** Let  $\mathfrak{G}$  first be a group without operators, and let  $\mathfrak{A}$  be its absolute automorphism domain, i.e. the system of all homomorphisms of  $\mathfrak{G}$  into itself.  $\mathfrak{A}$  is closed under

multiplication, since the product  $\sigma\tau$  is defined by

$$g(\sigma\tau) = (g\sigma)\tau, \quad g \in \mathfrak{G}, \quad \sigma, \tau \in \mathfrak{A}, \quad (1)$$

and plainly satisfies the associative law.

The group  $\mathfrak{G}$  becomes a group with operators when a set  $\mathfrak{B}$  of symbols  $O, H, \dots$  is given so that the products  $gO, gH, \dots$  are well-defined elements of  $\mathfrak{G}$  and generate homomorphisms of  $\mathfrak{G}$  into itself:

$$(gh)O = gO \cdot hO.$$

If the operator domain  $\mathfrak{B}$  is multiplicatively closed, then the induced map into the automorphism domain is multiplicatively homomorphic if and only if the associative relation

$$g(OH) = (gO)H, \quad g \in \mathfrak{G}, \quad O, H \in \mathfrak{B}, \quad (1a)$$

is satisfied. Correspondingly, a reciprocal homomorphism is obtained for left operators.

**2. Operator-homomorphic mapping.** If  $\mathfrak{G}$  is a group with operators, operator-homomorphy is defined by

$$(gO)\sigma = (g\sigma)O, \quad (2)$$

and, for left operators, by

$$(Og)\sigma = O(g\sigma). \quad (2^*)$$

For two operator domains  $\mathfrak{B}$  and  $\mathfrak{C}$ , the elements of  $\mathfrak{C}$  generate  $\mathfrak{B}$ -automorphisms, and at the same time the elements of  $\mathfrak{B}$  generate  $\mathfrak{C}$ -automorphisms, if and only if the domains are commutatively connected with  $\mathfrak{G}$ , respectively if the running associative law is fulfilled:

$$(gO)H = (gH)O, \quad (2a)$$

respectively

$$(OH)g = O(Hg). \quad (2a^*)$$

**3. Modules and bimodules over rings.** A right module  $\mathfrak{M}$  over a ring  $\mathfrak{R}$  is an additive abelian group with the elements of  $\mathfrak{R}$  as right operators, satisfying the associativity relation and the distributive relations

$$(g + h)\rho = g\rho + h\rho, \quad g(\rho + \sigma) = g\rho + g\sigma. \quad (3a)$$

The corresponding definitions hold for left modules.

There are two kinds of bimodules. Right modules over two rings  $\mathfrak{R}$  and  $\mathfrak{S}$  are called bimodules when, in addition to the individual module laws, one has

$$(m\rho)\sigma = (m\sigma)\rho.$$

An  $\mathfrak{R}$ -left,  $\mathfrak{S}$ -right module is a bimodule when the running associative law

$$\rho(m\sigma) = (\rho m)\sigma$$

is added.

If the operator domain  $\mathfrak{R}$  of an additively written abelian group  $\mathfrak{M}$  is a ring, then the map from  $\mathfrak{R}$  into the absolute automorphism ring is ring-homomorphic precisely when  $\mathfrak{M}$  is a right  $\mathfrak{R}$ -module; for left modules it is reciprocally ring-homomorphic. If  $\mathfrak{M}$  is a right module over  $\mathfrak{R}, \mathfrak{S}$ , then  $\mathfrak{M}$  is a bimodule precisely when  $\mathfrak{R}$  maps homomorphically into a subring of the  $\mathfrak{S}$ -automorphism ring and  $\mathfrak{S}$  maps homomorphically into a subring of the  $\mathfrak{R}$ -automorphism ring. For  $\mathfrak{R}$ -left,  $\mathfrak{S}$ -right bimodules, the  $\mathfrak{R}$ -map is reciprocal and the  $\mathfrak{S}$ -map direct.

In particular:

1) If  $\mathfrak{R}$  is a ring with identity, then  $\mathfrak{R}$  is directly isomorphic to its automorphism ring as a left  $\mathfrak{R}$ -module, and reciprocally isomorphic to its automorphism ring as a right  $\mathfrak{R}$ -module.

2) If  $\mathfrak{R}$  is a full matrix ring over a generally noncommutative field  $A$ ,

$$\mathfrak{R} = \sum c_{ik}A = \sum Ac_{ik},$$

then  $A$  is reciprocally isomorphic to the automorphism field of the simple right ideals, and directly isomorphic to the automorphism field of the simple left ideals.

**4. Passage from a right bimodule to a one-sided module over a product ring.** The significance of the right bimodule rests essentially on this passage.

**Theorem.** Let  $\mathfrak{M}$  be a right module over a ring  $\mathfrak{T}$  which contains two elementwise commuting subrings  $\mathfrak{R}$  and  $\mathfrak{S}$ . Then  $\mathfrak{M}$  may also be viewed as a bimodule over  $\mathfrak{R}$  and  $\mathfrak{S}$ . Conversely, if a bimodule over  $\mathfrak{R}$  and  $\mathfrak{S}$  is given, and if there exists a product ring  $\mathfrak{T}$  in which  $\mathfrak{R}$  and  $\mathfrak{S}$  commute elementwise, then  $\mathfrak{M}$  can also be regarded as a right module over  $\mathfrak{T}$ , with the  $\mathfrak{T}$ -operation extending the given  $\mathfrak{R}$ - and  $\mathfrak{S}$ -operations.

The product ring in the absolute automorphism ring consists of all elements of the form

$$\sum_i \rho_i \sigma_i + \rho + \sigma$$

(and if  $\mathfrak{R}, \mathfrak{S}$  have identities, the extra terms  $\rho, \sigma$  disappear). The homomorphisms

$$\mathfrak{R} \rightarrow \overline{\mathfrak{R}}, \quad \mathfrak{S} \rightarrow \overline{\mathfrak{S}}$$

extend by

$$\sum_i \rho_i \sigma_i + \rho + \sigma \mapsto \sum_i \bar{\rho}_i \bar{\sigma}_i + \bar{\rho} + \bar{\sigma}$$

to a homomorphism  $\mathfrak{T} \rightarrow \overline{\mathfrak{T}}$ . Hence

$$m\left(\sum_i \rho_i \sigma_i + \rho + \sigma\right) = \sum_i (m\rho_i) \sigma_i + m\rho + m\sigma$$

is well-defined and defines a  $\mathfrak{T}$ -module containing the given  $\mathfrak{R}, \mathfrak{S}$ -module structure.

## §2. Reciprocal and Direct Representation

**1. Modules of linear forms.** An  $\mathfrak{S}$ -right module  $\mathfrak{M}$ , where  $\mathfrak{S}$  is a ring with identity, is called a module of linear forms in  $\mathfrak{S}$  if  $\mathfrak{M}$  is a direct sum of  $n$  one-term  $\mathfrak{S}$ -modules,

$$\mathfrak{M} = m_1 \mathfrak{S} + \cdots + m_n \mathfrak{S},$$

such that  $m_i \mathfrak{S}$  is operator-isomorphic to  $\mathfrak{S}$ ; thus  $m_i s = 0$  always implies  $s = 0$ .

**Theorem 1.** If  $\mathfrak{M}$  is a module of linear forms in  $\mathfrak{S}$ , then the  $\mathfrak{S}$ -automorphism ring  $\mathfrak{U}$  of  $\mathfrak{M}$  is reciprocally isomorphic, not merely homomorphic, to the ring  $\overline{\mathfrak{U}}$  of all  $n$ -rowed matrices over  $\mathfrak{S}$ .

An arbitrary  $\mathfrak{S}$ -automorphism  $\alpha$  of  $\mathfrak{M}$  is completely described by

$$m_i \mapsto \bar{m}_i = m_i \alpha;$$

hence

$$\sum_i m_i s_i \mapsto \sum_i \bar{m}_i s_i = \sum_i (m_i \alpha) s_i = \sum_i (m_i s_i) \alpha.$$

Writing

$$(m_1 \alpha, \dots, m_n \alpha) = (m_1, \dots, m_n) A,$$

the correspondence  $\alpha \mapsto A$  is one-to-one, and

$$\alpha + \beta \mapsto A + B, \quad \alpha\beta \mapsto BA.$$

Thus the isomorphism is reciprocal. Consequently:

**Theorem 1'.** The automorphism ring of a module of linear forms of degree  $n$  in  $\mathfrak{S}$  admits an isomorphic, faithful reciprocal representation by the full matrix ring of  $n$ -rowed matrices over  $\mathfrak{S}$ .

**2. Representation and representation module.** A module of linear forms  $\mathfrak{M}$  in  $\mathfrak{S}$  is called a reciprocal representation module of  $\mathfrak{R}$  in  $\mathfrak{S}$  if  $\mathfrak{M}$  is a bimodule over  $\mathfrak{R}$  and  $\mathfrak{S}$  and at the same time a right module over both  $\mathfrak{R}$  and  $\mathfrak{S}$ . It is called a direct representation module if  $\mathfrak{M}$  is a bimodule and at the same time a left  $\mathfrak{R}$ -module and a right  $\mathfrak{S}$ -module.

**Theorem 2.** Every reciprocal, respectively direct, representation module generates a class of equivalent reciprocal, respectively direct, representations of  $\mathfrak{R}$  in  $\mathfrak{S}$ , and all such representations are obtained in this way.

For a reciprocal representation module,  $\mathfrak{R}$  maps directly homomorphically into a subring of the  $\mathfrak{S}$ -automorphism ring of  $\mathfrak{M}$ ; Theorem 1' then gives a reciprocal representation. Conversely, a reciprocal representation, composed with the reciprocal isomorphism into the  $\mathfrak{S}$ -automorphism ring, gives a direct homomorphism from  $\mathfrak{R}$ , hence a representation module. The direct case is obtained by reversing the homomorphism. Direct representations of  $\mathfrak{R}$  are reciprocal representations of the ring reciprocal to  $\mathfrak{R}$ , and conversely.

**3. Passage from a reciprocal representation module to a module over an extension ring.** For representation modules the transition theorem reads as follows.

If  $\mathfrak{M}$  is a right module over a ring  $\mathfrak{T}$  which contains two elementwise commuting subrings  $\mathfrak{R}$  and  $\mathfrak{S}$ , and if  $\mathfrak{M}$  is a module of linear forms over  $\mathfrak{S}$ , then  $\mathfrak{M}$  may also be regarded as a reciprocal representation module of  $\mathfrak{R}$  in  $\mathfrak{S}$ . Conversely, a reciprocal representation module may be regarded as a  $\mathfrak{T}$ -module whenever the product ring  $\mathfrak{T}$  exists.

If  $\mathfrak{M}$  is a reciprocal representation module of  $\mathfrak{R}$  in  $\mathfrak{S}$ , and also a  $\mathfrak{T}$ -module with  $\mathfrak{T}$  as product ring, then  $\mathfrak{R}$ ,  $\mathfrak{S}$ -isomorphism of  $\mathfrak{M}$  to a module  $\mathfrak{N}$  is equivalent to  $\mathfrak{T}$ -isomorphism. Thus each class of isomorphic  $\mathfrak{T}$ -modules corresponds to a reciprocal representation class of  $\mathfrak{R}$  in  $\mathfrak{S}$ , and conversely.

**Theorem on commuting matrices.** Let  $\mathfrak{R} \mapsto \mathfrak{R}^*$  be a reciprocal representation of degree  $n$  of  $\mathfrak{R}$  in  $\mathfrak{S}$ , and let  $\mathfrak{B}^*$  be the ring of all  $n$ -rowed matrices over  $\mathfrak{S}$  which commute elementwise with  $\mathfrak{R}^*$ . Then  $\mathfrak{B}^*$  gives a reciprocally isomorphic representation of the  $\mathfrak{R}$ ,  $\mathfrak{S}$ -automorphisms of the representation module generated by it, i.e. the  $\mathfrak{S}$ -automorphisms which are at the same time  $\mathfrak{R}$ -automorphisms; if the product ring  $\mathfrak{T}$  exists, these are the  $\mathfrak{T}$ -automorphisms.

The  $\mathfrak{R}$ ,  $\mathfrak{S}$ -automorphisms amount to a correspondence  $m_i \mapsto m'_i$  which produces the same representation. The elements  $m'_i$  need not form an  $\mathfrak{S}$ -basis of  $\mathfrak{M}$ ; correspondingly, the matrices in  $\mathfrak{B}^*$  need not be invertible. In this extended sense

$$(m'_1, m'_2, \dots, m'_n)a = (m'_1, m'_2, \dots, m'_n)A$$

remains correct, although  $A$  is no longer uniquely determined by  $a$ .

The diagonal matrices  $E \cdot s$  give a directly isomorphic representation of  $\mathfrak{S}$ , the identity representation of  $\mathfrak{S}$ . The correspondence from  $\mathfrak{T}$  to matrices in  $\mathfrak{S}$  defined by  $\mathfrak{R}$  and  $\mathfrak{S}$  is therefore not a representation of the ring  $\mathfrak{T}$ , but an extension of the reciprocal representation of  $\mathfrak{R}$  to one which is operator-homomorphic over  $\mathfrak{S}$ :

$$\sum_i r_i s_i \mapsto \sum_i R_i s_i.$$

In particular, the reciprocal representation of  $\mathfrak{R}$  is operator-homomorphic with respect to the intersection  $[\mathfrak{R}, \mathfrak{S}]$  of  $\mathfrak{R}$  and  $\mathfrak{S}$ , which lies in the center of  $\mathfrak{R}$ .

### §3. Modules with Respect to a Field

In what follows only representations in generally noncommutative fields will be involved. We therefore collect a few simple results on modules of linear forms over a field.

**1. Normal basis of a submodule with respect to a given basis of the full module.** Let  $A$  be a field and

$$\mathfrak{N} = x_1 A + \dots + x_n A$$

a module of linear forms of rank  $n$ . Let

$$\mathfrak{L} = z_1 A + \dots + z_\ell A$$

be a submodule of rank  $\ell \leq n$ . The  $z_i$  are called a normal basis with respect to the  $x_i$  if they are of the form

$$z_i = x_i - (x_{\ell+1}\alpha_{i,\ell+1} + \dots + x_n\alpha_{i,n}), \quad i = 1, \dots, \ell.$$

Every module  $\mathfrak{L}$  has, after a suitable numbering of the  $x_i$ , such a normal basis. Indeed,

$$\mathfrak{N} = \mathfrak{L} + x_{\ell+1}A + \cdots + x_nA,$$

and so

$$x_i \equiv x_{\ell+1}\alpha_{i,\ell+1} + \cdots + x_n\alpha_{i,n} \pmod{\mathfrak{L}}, \quad i = 1, \dots, \ell.$$

Thus the corresponding  $z_i$  lie in  $\mathfrak{L}$  and exhaust  $\mathfrak{L}$ .

**2. Extension module.** Let  $A$  again be a field, and let  $P$  be a subfield. An  $A$ -module  $N$  of rank  $n$  is called an extension module of a  $P$ -module  $M$ , written  $N = M_A$ , if  $M$  is a submodule of  $N$  of the same rank. If

$$M = y_1P + \cdots + y_nP,$$

then

$$M_A = y_1A + \cdots + y_nA.$$

Conversely, for every  $P$ -module  $M$ , the extension module exists uniquely up to module isomorphism.

**Corollary a).** If  $z_1, \dots, z_s$  are linearly independent over  $P$  in  $M$ , then they are also linearly independent over  $A$  in  $M_A$ . Thus every  $P$ -module

$$T = z_1P + \cdots + z_sP$$

inside  $M$  generates an extension module

$$T_A = z_1A + \cdots + z_sA \subset M_A.$$

**Corollary b).** Every  $P$ -module  $T$  inside  $M$  is a contraction module, i.e. the intersection of its extension module with  $M$ :

$$T = T_A \cap M.$$

**3. Theorem on invariant modules.** Let  $N = M_A$  be the extension module of a  $P$ -module  $M$ , and let  $P$  be the full invariant field for a group  $\mathfrak{G}$  of ring automorphisms of  $A$ . The group  $\mathfrak{G}$  is defined as an operator domain on  $N$  by

$$Gm = m, \quad m \in M,$$

and

$$G\left(\sum_i m_i \alpha_i\right) = \sum_i m_i G(\alpha_i).$$

Lemma:  $M$  consists precisely of those elements of  $N$  fixed by  $\mathfrak{G}$ .

**Theorem.** The extension modules  $L_A$  of submodules  $L$  of  $M$ , and only these, are admissible submodules with respect to  $\mathfrak{G}$  as operator domain.

One direction is clear. Conversely, let  $L$  be admissible for  $\mathfrak{G}$ , and let  $z_1, \dots, z_\ell$  be a normal basis of  $L$  with respect to a  $P$ -basis  $x_1, \dots, x_n$  of  $M$ . For every  $G \in \mathfrak{G}$ ,

$$G(z_i) = x_i - (x_{\ell+1}G(\alpha_{i,\ell+1}) + \cdots + x_nG(\alpha_{i,n}))$$

again lies in  $L$ . Comparing coefficients in  $x_1, \dots, x_\ell$  gives  $G(z_i) = z_i$ . Hence, by the lemma, the  $z_i$  lie in  $M$ , and  $L$  is the extension of the  $P$ -module

$$T = z_1P + \cdots + z_\ell P$$

in  $M$ , with  $T = L \cap M$ .

## §4. Hypercomplex Systems and Their Representation Classes

**1. Extension theorem.** We now combine the representation theorems of §2 with the module theorems of §3. Instead of arbitrary rings  $\mathfrak{R}$ , we consider hypercomplex systems with identity  $S, T, \dots$  over a commutative field  $P$ :

$$S = x_1P + \dots + x_nP.$$

Instead of arbitrary representation rings  $\mathfrak{S}$ , we consider only fields  $A, B, \dots$ , unless otherwise stated with center  $P$ . In this case there always exists the product ring in which  $S$  and  $A$  commute elementwise, with intersection  $P$ ; this is the direct product  $S \times A$ , which is at the same time the extension module  $S_A$  of  $S$ :

$$S_A = x_1A + \dots + x_nA.$$

It will therefore also be called the extension ring.

Under this specialization, the theorem on invariant modules becomes:

**Extension theorem.** Every two-sided ideal of  $S_A$  is the extension ideal of an ideal of  $S$ , namely the extension of its intersection with  $S$ .

Indeed, let the chosen group  $\mathfrak{G}$  be the group of inner automorphisms of  $A$ . Then  $P$ , as center of  $A$ , is the full invariant field, while the elementwise commutativity of  $A$  with  $S$  says that  $\mathfrak{G}$  is extended to  $S_A$  so that  $S$  is the full invariant domain. Every two-sided ideal  $\mathfrak{a}$  of  $S_A$  is an admissible subgroup, since  $\alpha^{-1}\mathfrak{a}\alpha \subseteq \mathfrak{a}$ , and so it is the extension of its intersection module  $\mathfrak{a} \cap S$ , which is an ideal in  $S$ .

Addendum: If  $S$  is commutative, then  $S$  is the center of  $S_A$ . More generally, the center of  $S$  is also the center of  $S_A$ . By  $A_r, B_n, \dots$  we shall always mean matrix rings of the indicated degree over  $A, B, \dots$ :

$$A_r = \sum A c_{ik} = \sum c_{ik} A.$$

We shall use essentially the extension theorem for simple systems:

If  $S$  is simple, then  $S_A$  is also simple:

$$S_A = \sum B c_{ik} = \sum c_{ik} B = B_t,$$

with associated field  $B$ , whose center contains  $P$ . Here  $A$ , and hence  $B$ , may have infinite rank over  $P$ ; only  $S$  is assumed hypercomplex. If  $S$  and  $A$  have common center  $P$ , then  $P$  is also the center of  $S_A$ . More generally, if  $S$  is a simple hypercomplex system, then the direct product  $S \times A_r$  is simple.

**2. Representation classes.** By a reciprocal or direct representation of a hypercomplex system we mean, as usual, a representation not equal to the zero representation and operator-homomorphic with respect to the coefficient field  $P$ .

**Theorem on representation classes.** Let  $S$  be hypercomplex with identity and coefficient field  $P$ , and let  $A$  be a field with center  $P$ . Then the number of distinct irreducible reciprocal representation classes of  $S$  in  $A$  is equal to the number of classes of simple right ideals in the quotient ring of  $S_A$  by its radical. If  $S$  is a simple system, then it has exactly one irreducible reciprocal and one irreducible direct representation class in  $A$ .

By the corollary to the transition theorem, the simple reciprocal representation classes and the simple module classes over  $S_A$  correspond one-to-one. Since  $S_A$  has finite rank over the field  $A$ , it satisfies the maximum and minimum conditions for one-sided ideals. If  $S$  is simple, then  $S_A$  is simple; hence there is only one simple one-sided ideal class, and so only one irreducible reciprocal representation class. The reciprocal system gives the corresponding assertion for direct representations.

**3. Rank relation.** If  $S$  is simple and

$$S_A = \sum B c_{ik}$$

has finite rank both over  $A$  and over  $B$ , then

$$n = rt, \quad n = (S : P), \quad t^2 = (S_A : B), \quad (1)$$

where  $r$  is the degree of the irreducible reciprocal representation.

**4. Sharpening of the rank relation when  $A$  has finite rank over  $P$ .** In this case the three relations are

$$(S : P) = n = rt, \quad (1)$$

$$(B : P)t = (A : P)r, \quad (2)$$

$$(S : P)(B : P) = (A : P)r^2. \quad (3)$$

Here (2) denotes the rank over  $P$  of a simple right ideal of  $S_A$ , expressed by the ranks over  $B$  and  $A$ ; (3) follows from (2) by multiplying by  $r$ , using (1).

## §5. Application of the Representation Theorems to Matrix Rings

The reciprocal representation of  $S$  in  $A$  considered in §4 can also be regarded as a direct homomorphism of  $S$  into a subring of  $A_r$ , where  $\bar{A}$  denotes the field reciprocally isomorphic to  $A$ . The principle of the following paragraphs is the converse one: to reduce the investigation of matrix rings  $A_r$  to representation theory in the reciprocally isomorphic field  $A$ .

**1. Reducible and irreducible embedding in  $A_r$ .** Let  $A$  and  $\bar{A}$  be reciprocally isomorphic fields, of finite or infinite rank over their center  $P$ . The simple system  $S$  over  $P$  is called irreducibly, respectively reducibly, embeddable in  $A_r$  if  $S$  has an irreducible, respectively reducible, reciprocal representation of degree  $r$  in  $A$ . If  $S$  is irreducibly embeddable in  $A_r$ , then it is reducibly embeddable in all matrix rings  $A_{rs}$ ,  $s > 1$ . The isomorphism always extends the identity of  $P$ ; by §4, 3,  $r$  divides the rank  $n$  of  $S$  over  $P$ .

In this way one obtains all simple subrings containing  $P$ , of finite rank over  $P$ , in the various  $A_r$ . For every such subring is reciprocally isomorphic to a subring of  $\bar{A}_r$  and therefore has a reducible or irreducible reciprocal representation in  $A$ .

**2. Theorem on inner automorphisms.** Let  $S^{(1)}$  and  $S^{(2)}$  be two simple subrings of  $A_f$  containing  $P$ , of finite rank over  $P$ . If  $S^{(1)}$  and  $S^{(2)}$  are isomorphic over  $P$ , then this isomorphism is induced by an inner automorphism of  $A_f$ .

Indeed,  $S^{(1)}$  and  $S^{(2)}$  give generally reducible embeddings of a simple system  $S$  in  $A_f$ , hence direct representations of degree  $f$  in  $A$ . They belong to the same reducible direct representation class; the transforming matrix induces the automorphism. If  $A$  itself has finite rank over  $P$ , so that it is hypercomplex, this becomes the familiar theorem: two simple subsystems of  $A_f$  containing the center  $P$ , and isomorphic over  $P$ , are carried into one another by an inner automorphism of  $A_f$ . In particular, every automorphism of  $A_f$  is inner.

**3. Theorem on elementwise commuting subrings.** Let  $S$  be a simple system contained in  $A_f$  and containing  $P$ , and let  $R$  be the totality of all elements of  $A_f$  which commute elementwise with  $S$ . Then  $R$  is also a matrix ring:

$$R = B_s.$$

The associated fields  $B$  of  $S_A$  and  $B$  of  $R$  are reciprocally isomorphic. The intersection of  $R$  and  $S$  is the center of  $S$ . The ring  $R$  is a field if and only if  $S$  is irreducibly embedded in  $A_f$ .

By §2, 3, the ring  $R$  is directly isomorphic to the automorphism ring of the reciprocal representation module  $\mathfrak{M}$  of  $S$  in  $A$ . This is the automorphism ring of  $\mathfrak{M}$ , regarded as an  $S_A$ -module. If  $\mathfrak{M}$  decomposes into  $s$  simple  $S_A$ -modules and one writes, as in §4, 1,

$$S_A = \sum Bc_{ik},$$

then  $R$  is isomorphic to  $\sum Bc_{ik}$ , where the two fields are reciprocally isomorphic. Thus  $R = B$  is a field precisely when  $s = 1$ , i.e. when  $S$  is irreducibly embedded. By definition,  $R \cap S$  is the center of  $S$ .

**4. Sharpened commutation theorem for hypercomplex matrix rings.** In this case the rank relations of §4, 4 give the sharpening:

The simple subsystems of  $A_f$  which contain  $P$ , where  $A$  has finite rank over its center  $P$ , split into pairs  $S, \bar{S}$ , such that  $\bar{S}$  consists exactly of all elements which commute elementwise with  $S$ , and conversely. The associated fields of  $S_A$  and  $\bar{S}$  are reciprocally isomorphic, as are those of  $S$  and  $\bar{S}_A$ . The intersection of  $S$

and  $\bar{S}$  is the common center. One subring is a field if and only if the other is irreducibly embedded. The product of the ranks over  $P$  of  $S$  and  $\bar{S}$  is the rank of  $A_f$ .

If  $f = rs = r's'$ , where  $S$  is irreducibly embeddable in  $A_r$  and  $\bar{S}$  in  $A_{r'}$ , then the associated fields of  $S$  and  $\bar{S}$  become isomorphic to a pair of commuting subrings of  $A_g$ , with  $g = ss'$ .

The product assertion follows directly from the rank relation. If  $S$  is irreducibly embedded, then  $\bar{S}$  is a field and is reciprocally isomorphic to  $B$ , where  $S_A = B_t$ ; the rank relation (3) of §4, 4 gives the assertion. If  $S$  is reducibly embedded, so  $\bar{S} = B_s$ , then  $f = rs$ , and multiplying (3) by  $s^2$  gives the assertion. This also shows that  $\bar{S}$  is precisely the totality of the elements commuting elementwise with  $S$ , and the remaining assertions follow from the theorem on commuting subrings.